



System Manual

Complete Operations Platform Guide

Web Command Portal · Supervisor Portal · Mobile Field Clients



Document Titan Protection System Manual

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Live system <https://titanprotection.org>

Built to Protect

Developed by Arch Luviah Technologies



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1. About Titan Protection

Titan Protection is an integrated security operations platform designed for private security companies, corporate protection teams, and institutional guard services. The system unifies command-and-control, field operations, geographic site management, personnel administration, and real-time incident response in a single live ecosystem.

This manual documents the complete Titan Protection system — not a single application or user role, but the full platform: web command portals, supervisor workspaces, Android field clients, shared database, and operational workflows that connect them.

1.1 System Purpose

- Register and geolocate protected premises and patrol points.
- Organise operations by territory with dedicated area supervisors.
- Deploy and monitor field personnel with GPS-verified attendance.
- Track patrol activity, incidents, and compliance in real time.
- Respond to distress (SOS) signals and operational alerts immediately.
- Maintain audit trails through the Occurrence Book and Data Explorer.
- Distribute mobile clients and keep all endpoints synchronised.

1.2 Who Uses This Manual

Audience	Interest
Master Administrators	Full platform setup, configuration, and governance
Operations Managers	Daily monitoring via Command Centre and GIS map
Area Supervisors	Territory-scoped web and mobile supervisor tools
IT / Deployment Teams	APK distribution, login configuration, database sync
Training & Compliance Officers	Workflows, checklists, audit records

2. System Architecture & Components

Titan Protection consists of four client applications sharing one API and database. All components read from and write to the same operational datastore when connected to the live server (Supabase).

Component	Platform	Primary function
Master Admin Web Portal	Browser — /	Full system administration and monitoring

Supervisor Web Portal	Browser — /supervisor	Territory-scoped operations dashboard
Titan Monitor	Android — com.titan.monitor	Field operations client for on-site personnel
Titan Supervisor	Android — com.titan.supervisor	Field client for supervisors — GPS site capture, team view

2.1 Shared Infrastructure

- API layer — /api/state handles reads and operational mutations for all clients.
- Authentication — separate admin (email/password) and supervisor (6-digit PIN) sessions.
- Database — Supabase PostgreSQL; demo mode uses in-memory store when server unreachable.
- Real-time polling — web dashboards refresh every 10 seconds; mobile clients poll on comparable intervals.
- Production URL — <https://titanprotection.org> (Vercel deployment, web root: web/).

2.2 Recommended Deployment Sequence

1. Configure Master Admin credentials and connect to live database.
2. Create territories and register supervisors; deliver PINs.
3. Register premises with on-site GPS capture (± 5 m accuracy).
4. Add patrol places at each premises.
5. Register field personnel; assign one supervisor and premises per person.
6. Schedule shifts and distribute Titan Monitor APK.
7. Supervisors install Titan Supervisor APK for field GPS capture.
8. Monitor operations in Command Centre and GIS map.

3. User Roles & Access Model

Role	Portal / App	Scope
Master Admin	Web — /	Entire tenant — all territories, settings, data
Area Supervisor	Web — /supervisor + Titan Supervisor app	Assigned territories only
Field Officer	Titan Monitor app	Assigned premises and shifts

3.1 Supervisor Territory Scoping

Supervisors (web and mobile) see only data linked to their assigned territory IDs: premises in those territories, places under those premises, guards assigned to them or their territories, related shifts, attendance, alerts, and occurrence book entries.

Access rule: Supervisors cannot create territories, manage other supervisors, reassign guards to

another supervisor, or access Master Admin tools.

4. Data Flow & Synchronisation

Typical operational data flow across the system:

9. Admin or supervisor registers a premises with GPS → stored in database → appears on GIS map (green marker).
10. Patrol places added at premises → small red markers on map; checkpoint records sync for mobile patrol.
11. Field officer clocks in via Titan Monitor → attendance record → Command Centre and map update.
12. Incident or SOS logged on mobile → Occurrence Book entry → alert banner on web dashboards.
13. Supervisor captures site GPS on Titan Supervisor app → premises/places update via API.
14. All web dashboards poll /api/state; changes propagate within seconds on live connection.

Connection indicator: Server Connected / Live = production database. Demo Mode = in-memory only; data may not persist.

PART II · ACCESS & WEB PORTALS

5. Signing In

5.1 Login Chooser — /login

Welcome to Titan Protection — choose your role:

Role card	Description	Action
Master Admin	Full access — supervisors, territories, personnel, premises, settings	Admin sign in →
Supervisor	Area-scoped — command centre, personnel, premises in assigned territories	Supervisor sign in →

Footer: Download Titan Monitor & Supervisor APKs → /downloads

5.2 Master Admin — /admin/login

15. Enter email and password configured by your system administrator.
16. Click Sign In → redirected to / (Master Admin Dashboard).



17. Sign Out from the page header when finished.

5.3 Supervisor — /supervisor/login

18. Enter 6-digit PIN on the numeric keypad (same PIN as Titan Supervisor mobile app).

19. Redirected to /supervisor (Supervisor Dashboard).

20. Forgotten PIN: Master Admin resets from Supervisor & Territories → Reset PIN.

6. Master Admin Web Portal

Full operations hub. Sidebar navigation:

Module	Function
Supervisor & Territories	Territories, supervisor accounts, WhatsApp setup
Guard Management	Personnel profiles, shifts, attendance, alerts, performance
Register Premises	Protected sites, patrol places, GPS capture
GIS Operations Map	Live geographic operations view
Command Centre	Real-time stats, map, Occurrence Book, visitors
Data Explorer	Database collections — search, export, edit
Master Admin	Sync, checklists, audits, data management
Mobile App Downloads	APK distribution page

6.1 System Settings (sidebar)

Setting group	Options
Alerts & Sirens	Siren on critical alerts (SOS, critical guard alerts)
GPS & Geofencing	Geofence radius 5–8 m (default 6 m)
Patrol Monitoring	No-movement threshold 15–90 min (default 45 min)
License Compliance	Expiry warning window 14–90 days (default 60 days)

6.2 Connection & Refresh

Header badge shows Server Connected or Demo Mode Active. Dashboard polls every 10 seconds. Critical alerts may trigger audible siren when enabled.

7. Supervisor Web Portal

Territory-scoped mirror of key admin modules:

Module	Supervisor scope
Register Premises	Premises in assigned territories
Guard Management	Scoped personnel only

GIS Operations Map	Scoped sites, places, events
Command Centre	Scoped live monitoring
Data Explorer	Scoped records; limited edit tables

Not available to supervisors: Supervisor & Territories admin, Master Admin tab, System Settings, Mobile Downloads link, Reset Patrol, cross-territory data.

PART III · MOBILE FIELD CLIENTS

8. Titan Monitor — Field Operations Client

Package: com.titan.monitor · Android APK from /downloads

Purpose: Primary field client for on-site security personnel.

8.1 Sign-In & Profile

- 6-digit PIN login (delivered by email on registration).
- First-login PIN change flow when required.
- Profile photo upload; assigned premises resolved from guard profile and active shift.
- Over-the-air update prompt when newer APK published.

8.2 Main Modules (bottom navigation)

Tab	Function
Patrol	Clock-in/out with GPS geofence verification; patrol point progress; movement tracking
Incidents	Log incidents with photo and voice memo attachments; syncs to Occurrence Book
Checklists	Complete published audit/checklist templates from web admin
Access	Visitor QR scanning and access desk functions
Profile	Guard details, documents, performance summary, sign out

8.3 Key Field Operations

- GPS clock-in — must be within premises geofence radius (system setting, default 6 m).
- Patrol progress — tracks scanned/completed patrol points for assigned site.
- SOS distress — triggers CRITICAL DISTRESS SIGNAL on all web Command Centres.
- No-movement monitoring — server-side alerts if guard stationary beyond threshold.
- Geofence exit alerts — triggered when on-duty guard leaves site boundary.

9. Titan Supervisor — Mobile Client

Package: com.titan.supervisor · Android APK from /downloads

Purpose: Field tool for area supervisors — complements the supervisor web portal.

9.1 Sign-In

Same 6-digit PIN as supervisor web portal. Session scoped to supervisor ID and assigned territories.

9.2 Main Modules

Tab	Function
Home	Dashboard summary — territories, premises, guards, alerts
Sites	Register premises and patrol places with high-accuracy GPS capture (± 5 m target)
Team	View and manage scoped guard team
Profile	Supervisor photo, PIN change, app update, sign out

9.3 GPS Capture

Sites panel uses extended GPS warm-up and cluster-median filtering for accurate coordinates. Save is optimised for fast completion after GPS lock. Captured coordinates sync to web GIS map.

PART IV · CORE OPERATIONAL MODULES

10. Territory & Supervisor Administration

Master Admin only — Supervisor & Territories module.

10.1 Territories

21. Add Territory → name, city, description, suburbs list.
22. Territories scope all premises, supervisors, and filtered data views.
23. Delete only after reassigning linked premises and personnel.

10.2 Supervisors

24. Add Supervisor → name, role, phone, email, assigned territories.
25. System generates 6-digit PIN; delivered via email + WhatsApp manual send.
26. Reset PIN, upload photo, edit status (Active / Suspended / Off Duty).

10.3 WhatsApp Integration

Manual WhatsApp model — system opens WhatsApp with pre-filled message; user taps Send. Used for PIN delivery backup, shift notifications, and supervisor-to-guard messaging.

11. Premises & Site Registration

Available on Master Admin and Supervisor web portals (supervisor: scoped territories). Also via Titan Supervisor mobile Sites tab.

11.1 Register Premises

- 27. Premises name, owner/client, address, territory, city, suburb.
- 28. Capture GPS on site — target ± 5 m accuracy.
- 29. Share Premises ID with field personnel for Titan Monitor site linking.

11.2 Patrol Places

- 30. Add Place under selected premises — name, type, schedule, description.
- 31. Capture place GPS at physical location.
- 32. Map display: small red P markers. NFC tagging deferred for future release.

11.3 Map Markers (current)

Marker	Colour / size	Meaning
Premises	Green — site initials	Registered protected site centre
Patrol place	Red — small P pin	Important location inside a premises

12. Personnel & Shift Management

Guard Management module — personnel administration (web).

Sub-tab	Purpose
Guard Profiles	Register personnel, documents, bulk supervisor assignment
Shift Roster	Schedule shifts; WhatsApp notification on save
Live Attendance	On-duty status, geofence/movement badges, history
Supervisor Alerts	Movement, geofence, license alerts — dismiss when resolved
Performance	Composite scorecards
Shift Swaps	Approve/reject mobile swap requests
Messaging	Email and WhatsApp configuration and send

12.1 Personnel Registration Rules

- Each field officer assigned to exactly one supervisor (supervisorId).
- Supervisors registering new personnel must assign them to themselves.
- Bulk tools: Assign all to supervisor; Auto-match by territory.
- PIN delivered via email for Titan Monitor login.

13. Command Centre & Real-Time Monitoring

Shared by Master Admin and Supervisor portals (supervisor: scoped data).

Panel	Content
SOS banner	CRITICAL DISTRESS SIGNAL — Dismiss Alarm
Statistics	Compliance rate, taps scanned, active guests, guards on duty
Live map	Compact GIS view of sites and patrol points
Occurrence Book	Chronological field events — status workflow, photo/voice attachments
Visitor registry	Check-in form and live visitor log table

14. GIS Operations Map

Full-screen map (Leaflet) with sidebar in dedicated tab; compact embed in Command Centre.

14.1 Basemaps

- Google Satellite Hybrid (default), Google Streets, Google Terrain, Dark Matter GIS, Esri World Imagery.

14.2 View Presets

Preset	Use case
Operations	Live personnel, geofences, trails, alerts
Patrol	Premises and patrol places only (simplified default)
Emergency	Alerts, heatmap, activity focus
Overview	Sites and places across territories

14.3 Layer Toggles

- Protected premises, patrol places, geofences, GPS accuracy rings, patrol routes.
- Live personnel, movement trails, shift roster, alerts, activity, heatmap, territory zones.
- NFC checkpoint layer hidden until NFC patrol map integration is enabled.

14.4 Map Tools

- Search, zoom, fit all sites, measure distance, copy GPS from popups, fly-to on SOS.

15. Data Explorer & Records Management

Browse all system collections — Core, Geography, People, Operations, Communications, Compliance, Emergency categories.

- Search, paginate (30 rows), copy JSON, export CSV/JSON.
- Inline edit (admin: all editable tables; supervisor: premises, places, guards, shifts, alerts, OB).
- Sensitive fields redacted — PINs, passwords, raw GPS trails, attachments.

PART V · PLATFORM SERVICES

16. Communications, PIN Delivery & Alerts

Event	Email	WhatsApp
New field officer registered	Titan Monitor PIN	Manual send modal
New supervisor registered	Supervisor PIN	Manual send modal
PIN reset	New PIN emailed	WhatsApp opens
Shift scheduled	—	Notification message
SOS triggered	—	Command Centre banner + optional siren

16.1 Alert Types

- SOS panic — highest priority; red banner all dashboards.
- No-movement — field officer stationary beyond threshold.
- Geofence exit — on-duty officer left site boundary.
- License expiry — compliance warning window.
- Shift swap request — pending supervisor action.

17. System Settings & Master Administration

17.1 Master Admin Tab

- Reload from database — sync from Supabase.
- Clear all data — destructive demo reset (production caution).
- Checklist Template Builder — publish audit forms for Titan Monitor.
- Completed Audits — view submitted checklist responses.
- System Configuration summary — read-only settings display.

17.2 Security

- Role-isolated sessions — admin vs supervisor cookies.
- PIN redaction in Data Explorer and supervisor API responses.
- Supervisor mutation guard — server validates territory scope on every action.
- HTTPS enforced in production.

18. Mobile App Distribution & Updates

Public page: /downloads

App	Package	Audience
Titan Monitor	com.titan.monitor	Field operations personnel
Titan Supervisor	com.titan.supervisor	Area supervisors

18.1 Installation

33. Download APK from /downloads.
34. Allow installation from unknown sources (Android).
35. Sign in with PIN.
36. Use in-app Update app when newer build published — OTA without manual reinstall.

PART VI · SUPPORT

19. Troubleshooting & FAQ

Issue	Resolution
Demo Mode showing	Verify Supabase connection; Reload from database
Wrong site on Titan Monitor	Verify personnel premises assignment; logout/login after app update
0/0 patrol points	Ensure places exist under premises; checkpoints sync from places
GPS capture fails	Enable location; stand outdoors; wait for ± 5 m lock
Blue NFC markers on map	Disable checkpoints layer; NFC map integration deferred
Supervisor PIN failed	Master Admin → Reset PIN
Data not updating	Wait 10 s auto-refresh or use Refresh in Data Explorer

19.1 FAQ

Question	Answer
Is this a guard-only manual?	No — this documents the entire Titan Protection system for all roles.
Can one officer have two supervisors?	No — exactly one supervisor per field officer.
Do supervisors see all company data?	No — only assigned territories.
How many map marker types now?	Two — green premises, red patrol places.
Where are APKs?	/downloads or login page footer link.

Appendix A — Quick Reference

Resource	URL / path
Login chooser	/login
Admin dashboard	/
Supervisor dashboard	/supervisor
Mobile downloads	/downloads
Production	https://titanprotection.org

System Onboarding Checklist

37. Connect live database; confirm Server Connected.
38. Create territories and supervisors.
39. Register premises with GPS.
40. Add patrol places.
41. Register personnel; assign supervisor and premises.
42. Schedule shifts.
43. Deploy Titan Monitor and Titan Supervisor APKs.
44. Verify Command Centre and GIS map.

Appendix B — Glossary

Term	Definition
Titan Protection System	The complete platform — web portals, mobile apps, API, and database
Premises	A registered protected site with GPS coordinates
Patrol place	Important location inside a premises (red map marker)
Territory	Geographic operating area (city + suburbs)
Command Centre	Real-time web monitoring module
Occurrence Book (OB)	System-wide chronological event log
Titan Monitor	Android field operations client
Titan Supervisor	Android supervisor field client
Geofence	GPS radius for clock-in verification (5–8 m)



Compliance rate	Patrol checkpoint scan completion percentage
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